

Smoothness-Enhanced Pruning Strategies for Manifold-Adaptive Empirical Cubature Methods

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Abstract

Empirical cubature and hyper-reduction techniques rely on the construction of sparse integration rules that reproduce, with high accuracy, the integral contributions of high-dimensional models over reduced spaces. In recent work (see MAW_ECM.pdf), a **manifold-adaptive weighted empirical cubature** strategy was introduced, enabling the **progressive pruning of integration points** while **preserving exactness conditions** at discrete sampling locations on a low-dimensional manifold. The method relies on **local least-norm reconstructions** of the cubature weights and exhibits excellent performance in early stages of the reduction process.

However, **as the pruning advances, positivity constraints on the weights become active**, leading to local degeneracies and **loss of smoothness of the weight field across the manifold**. This work introduces a smoothness-enhanced extension of the existing pruning framework, designed to **mitigate spurious local discontinuities** without resorting to fully global constrained optimization. The proposed strategy preserves the original algorithmic structure, while incorporating **global graph-based smoothing operators** and lightweight positivity restoration steps. The resulting method maintains exactness, non-negativity, and continuity of the weights over the manifold, while remaining computationally efficient and well-suited for large numbers of manifold sampling points.

1 Introduction

Empirical cubature methods have become a central tool in the development of hyper-reduced models for large-scale nonlinear problems, particularly in the context of reduced-order modeling and multiscale finite element formulations. By selecting a **small subset of integration points and associated weights**, these methods enable the efficient evaluation of internal forces, residuals, or energy contributions while preserving the accuracy of the original high-fidelity model.

In previous work, a manifold-adaptive weighted empirical cubature method was proposed, combining ideas from empirical cubature, subspace adaptation, and **continuous pruning** strategies inspired by cubature-based reduction techniques. The method operates by **progressively eliminating integration points and redistributing their contribution among the remaining ones**, while **enforcing exactness constraints** at a discrete set of sampling points lying on a low-dimensional manifold of reduced or latent coordinates. At each stage of the **pruning process**, the **cubature weights are reconstructed locally by solving underdetermined systems of linear equations**, selecting the **least-norm solution that satisfies the imposed exactness and volume constraints**.

This local reconstruction strategy is highly effective during the initial stages of the reduction process. In particular, when the **least-norm solution naturally yields non-negative weights**, the resulting cubature rule **remains smooth across the manifold** and exhibits excellent stability properties.

□ [JAHO: *Stability properties? Why ? I don't think this fits here...*] □

However, as the pruning progresses and the number of remaining integration points decreases, the **least-norm reconstruction may produce negative weights**. Enforcing non-negativity then requires the activation of **inequality constraints**, which in turn can induce abrupt changes in the local support of the cubature rule and lead to **spatially incoherent weight patterns across neighboring manifold samples**.

The objective of the present work is not to **replace the original pruning methodology**, nor to reformulate it as a fully global constrained optimization problem. Instead, the goal is to introduce a **minimal and computationally efficient enhancement that restores smoothness of the cubature weights** over the manifold when **positivity constraints become active**. The proposed approach preserves the original local reconstruction philosophy, while complementing it with global smoothing operators derived from the graph structure of the manifold sampling points. Positivity is enforced through simple local corrections, and exactness constraints are reimposed iteratively, resulting in a robust and scalable algorithm suitable for large manifolds and high-dimensional integration rules.

In the following sections, we describe the smoothness-enhanced pruning strategy in detail, discuss its algorithmic implementation, and analyze its effect on the stability and continuity of the resulting cubature rules.

2 Manifold-Adaptive Cubature as a Sparsification Problem

The formulation of empirical cubature as a sparsification problem follows the optimization-based viewpoint introduced in the Subspace Adaptive Weights Empirical Cubature Method (SAW-ECM). In that framework, sparsity is not imposed on individual weight vectors associated with different subspaces, but rather on the *union of their supports*, achieved by minimizing the zero-norm of the sum of all weight vectors. This construction ensures that the same reduced set of integration points is shared across all subspaces.

The present work extends this idea from a finite collection of reduced subspaces to a continuous low-dimensional manifold of sampling points, while preserving the same sparsification principle.

2.1 SAW-ECM sparsification principle

Let $\mathbf{W} \in \mathbb{R}^{N_g}$ denote the full finite element quadrature weight vector associated with the high-fidelity discretization, and let $\boldsymbol{\zeta}^{(i)} \in \mathbb{R}^{N_g}$ denote the cubature weight vector associated with the i -th reduced subspace, $i = 1, \dots, k$. The design variables of the problem are the collection of vectors

$$\boldsymbol{\zeta}^{(1)}, \boldsymbol{\zeta}^{(2)}, \dots, \boldsymbol{\zeta}^{(k)},$$

each having the same dimension as $\mathbf{W}$.

Since the cubature weights are constrained to be nonnegative, the total number of selected integration points is given by the cardinality of the support of the sum of all weights,

$$m = \text{card}(\mathcal{E}) = \left\| \sum_{i=1}^k \boldsymbol{\zeta}^{(i)} \right\|_0,$$

that is, the number of positive entries of the aggregated weight vector.

Given a set of matrices $\mathbf{U}^{(i)} \in \mathbb{R}^{M \times N_g}$ representing the functions to be integrated exactly over each subspace, and right-hand sides $\mathbf{b}^{(i)} = \mathbf{U}^{(i)T} \mathbf{W}$, the SAW-ECM sparsification problem is stated

as

$$\begin{aligned} & \min_{\boldsymbol{\zeta}^{(1)}, \dots, \boldsymbol{\zeta}^{(k)}} \left\| \sum_{i=1}^k \boldsymbol{\zeta}^{(i)} \right\|_0 \\ & \text{subject to } \mathbf{U}^{(i)T} \boldsymbol{\zeta}^{(i)} = \mathbf{b}^{(i)}, \quad i = 1, \dots, k, \\ & \quad \boldsymbol{\zeta}^{(i)} \geq \mathbf{0}, \quad i = 1, \dots, k. \end{aligned} \tag{1}$$

This formulation promotes a common sparse support across all subspaces, while allowing the weights themselves to adapt independently.

2.2 Adaptation to a manifold of sampling points

In the manifold-adaptive setting considered in this work, the reduced model is characterized by a continuous low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$, parameterized by a vector of latent (or master) coordinates $\boldsymbol{\xi}$. The manifold is sampled at a finite set of points

$$\{\boldsymbol{\xi}^{(j)}\}_{j=1}^{N_m},$$

which play a role analogous to the subspaces in the SAW–ECM formulation.

For each sampling point $\boldsymbol{\xi}^{(j)}$, a cubature weight vector

$$\boldsymbol{\zeta}^{(j)} \in \mathbb{R}^{N_g}$$

is introduced, having the same dimension as the full quadrature weight vector. The collection of all such vectors constitutes the design variables of the manifold-adaptive sparsification problem.

At each sampling point, a set of linear exactness constraints is imposed. Denoting by $\mathbf{U}^{(j)} \in \mathbb{R}^{n_c \times N_g}$ the matrix of constraint functions evaluated at $\boldsymbol{\xi}^{(j)}$, and by $\mathbf{b}^{(j)} \in \mathbb{R}^{n_c}$ the corresponding exact integrals, the constraints read

$$\mathbf{U}^{(j)T} \boldsymbol{\zeta}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m. \tag{2}$$

Among these constraints, one enforces conservation of volume,

$$\sum_{i=1}^{N_g} \zeta_i^{(j)} = V, \quad \forall j, \tag{3}$$

where V denotes the total integration volume.

In direct analogy with SAW–ECM, sparsity is promoted by minimizing the cardinality of the support of the sum of all weight vectors over the manifold,

$$\mathcal{E} = \sum_{j=1}^{N_m} \boldsymbol{\zeta}^{(j)}, \tag{4}$$

leading to the manifold-adaptive sparsification problem

$$\begin{aligned} & \min_{\boldsymbol{\zeta}^{(1)}, \dots, \boldsymbol{\zeta}^{(N_m)}} \left\| \sum_{j=1}^{N_m} \boldsymbol{\zeta}^{(j)} \right\|_0 \\ & \text{subject to } \mathbf{U}^{(j)T} \boldsymbol{\zeta}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\ & \quad \boldsymbol{\zeta}^{(j)} \geq \mathbf{0}, \quad j = 1, \dots, N_m. \end{aligned} \tag{5}$$

This formulation enforces a common sparse set of integration points across the entire manifold, while allowing the cubature weights to vary with the latent coordinates.

□ **[JAH0:** *The preceding equation requires an additional term in the objective function (that will account for the “smoothness”)*] □

2.3 Initial cubature rule

The sparsification process is initialized using a reference cubature rule obtained from the standard Empirical Cubature Method. This initial rule provides a set of integration points and a weight vector $\bar{\zeta} \in \mathbb{R}^{N_g}$ that satisfies all exactness constraints. The initial manifold-adaptive weights are defined as

$$\zeta^{(j)} = \bar{\zeta}, \quad j = 1, \dots, N_m, \quad (6)$$

thus ensuring feasibility of the constraints at all manifold sampling points prior to pruning.

3 Dimensional Reduction of the Design Variables and Smoothness Requirements

3.1 Restriction to the initial ECM support

Rather than working at the level of the full-order quadrature weight vector associated with all finite element integration points, the present formulation operates on a reduced set of candidate points provided by an initial Empirical Cubature Method (ECM) rule. Let

$$\mathcal{I}_0 \subset \{1, \dots, N_g\}$$

denote the index set of integration points selected by the initial ECM procedure, with cardinality $N_0 = |\mathcal{I}_0| \ll N_g$. All subsequent weight vectors are defined only on this reduced set.

Accordingly, the manifold-adaptive weight vectors are redefined as

$$\zeta^{(j)} \in \mathbb{R}^{N_0}, \quad j = 1, \dots, N_m,$$

where each component corresponds to one of the integration points in $\mathcal{I}_0$. This restriction significantly reduces the dimensionality of the design space and reflects the practical implementation of the pruning strategy, in which integration points are progressively eliminated from the initial ECM rule.

All exactness and volume constraints introduced in the previous section are understood to be expressed with respect to this reduced set of candidate integration points.

3.2 Continuity requirements in MAW–ECM versus SAW–ECM

A fundamental distinction between the Subspace Adaptive Weights ECM (SAW–ECM) and the Manifold Adaptive Weights ECM (MAW–ECM) lies in the role of continuity of the cubature weights.

In SAW–ECM, the weight vectors associated with different reduced subspaces are independent, and no continuity requirement is imposed across subspaces. This is consistent with the fact that **subspaces represent discrete regimes**, and the resulting cubature rule is **not intended to be interpolated or evaluated continuously across them**.

In contrast, MAW–ECM is designed to produce a family of weight vectors

$$\zeta^{(j)}, \quad j = 1, \dots, N_m,$$

defined over a set of sampling points on a **low-dimensional manifold of latent coordinates**. The ultimate objective of the method is not only to obtain feasible cubature weights at these sampling points, but also to construct a **continuous mapping between the latent coordinates and the cubature weights through regression** or interpolation techniques.

For this reason, **continuity and smoothness of the weight vectors over the manifold are essential requirements in MAW-ECM**. Discontinuous or highly oscillatory weight fields would lead to ill-conditioned regression problems and nonphysical interpolation artifacts.

□ [JAHO: *Not exactly, what happens is that this is intended to be used in a HROM, and this places continuity requirements of the Jacobian matrix*] □

Consequently, the **sparsification problem introduced previously must be augmented with an additional mechanism that penalizes lack of smoothness** of the weights across neighboring manifold samples.

3.3 Graph-based smoothness penalty on a rectangular manifold grid

For clarity of exposition, we restrict attention to the case in which the manifold is parameterized by two latent variables,

$$\boldsymbol{\xi} = (\xi_1, \xi_2) \in \mathbb{R}^2,$$

and the sampling points $\{\boldsymbol{\xi}^{(j)}\}_{j=1}^{N_m}$ form a rectangular grid in the latent space. In this setting, the manifold samples can be naturally interpreted as the nodes of a structured finite element mesh.

A graph $\mathcal{G}$ is **constructed by connecting each sampling point to its nearest neighbors in the grid**. The corresponding **graph Laplacian can be interpreted as the conductance (or stiffness) matrix of a finite element discretization of a diffusion operator on the latent domain**. Denoting this matrix by

$$\mathbf{K} \in \mathbb{R}^{N_m \times N_m},$$

$\mathbf{K}$ is **sparse, symmetric, and positive semidefinite, and encodes the adjacency and connectivity of the manifold sampling points**.

□ [JAHO: *What is adjacency ?*] □

Smoothness of the cubature weights over the manifold is **promoted by penalizing variations of each weight component across neighboring sampling points**. Collecting the weights corresponding to a fixed integration point index $i \in \mathcal{I}_0$ into a vector

$$\boldsymbol{\zeta}_i = \begin{bmatrix} \zeta_i^{(1)} & \zeta_i^{(2)} & \cdots & \zeta_i^{(N_m)} \end{bmatrix}^T,$$

a natural smoothness measure is given by the quadratic form

$$\boldsymbol{\zeta}_i^T \mathbf{K} \boldsymbol{\zeta}_i.$$

Summing over all integration points in the reduced candidate set leads to the global smoothness penalty

$$\sum_{i \in \mathcal{I}_0} \boldsymbol{\zeta}_i^T \mathbf{K} \boldsymbol{\zeta}_i.$$

3.4 Smoothness-augmented manifold-adaptive sparsification problem

Taking into account the need for smoothness, the **manifold-adaptive sparsification problem introduced previously is augmented by a regularization term penalizing lack of continuity of the cubature weights over the manifold**. The resulting optimization problem

reads

$$\begin{aligned}
& \min_{\zeta^{(1)}, \dots, \zeta^{(N_m)}} \left\| \sum_{j=1}^{N_m} \zeta^{(j)} \right\|_0 + \frac{\alpha}{2} \sum_{i \in \mathcal{I}_0} \zeta_i^T \mathbf{K} \zeta_i \\
& \text{subject to } \mathbf{U}^{(j)T} \zeta^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\
& \quad \zeta^{(j)} \geq \mathbf{0}, \quad j = 1, \dots, N_m,
\end{aligned} \tag{7}$$

where $\alpha > 0$ is a regularization parameter controlling the trade-off between sparsity and smoothness.

This formulation preserves the sparsification principle of SAW–ECM, while incorporating an explicit mechanism to enforce continuity of the cubature weights across the manifold, which is essential for the subsequent regression-based construction of a continuous mapping between latent coordinates and cubature weights.

4 Intractability of the Sparsification Problem and Motivation for a Pruning Strategy

4.1 Intractability of the zero-norm sparsification problem

The **smoothness-augmented manifold-adaptive sparsification** problem introduced in the previous section involves the minimization of a **pseudo zero-norm objective** under linear equality constraints, non-negativity constraints, and an additional smoothness regularization term. Problems of this type are well known to be **computationally intractable** in general.

As discussed in the context of the SAW–ECM formulation, the **zero-norm objective is combinatorial in nature**, as it effectively requires the **identification of the smallest subset of integration points** whose associated weights satisfy all imposed constraints. Even in the absence of smoothness requirements, this class of problems is **NP-hard**, and exact solutions are only feasible for very small problem sizes.

□ [JAHO: *I don't think this is relevant (the feasibility for small problem sizes)*] □

The introduction of multiple weight vectors—one per manifold sampling point—together with continuity requirements further **exacerbates the complexity**, rendering direct optimization approaches **impractical** for the problem sizes of interest.

□ [JAHO: *No problem can be effectively attacked with this direct approach !*] □

In the present setting, the dimensionality of the design space scales with the product of the number of manifold sampling points and the number of candidate integration points retained from the initial ECM rule.

□ [JAHO: *Are you sure of that ? You have to consider the equality constraints as well.*] □

As a consequence, any direct attempt to solve the sparsification problem posed in the previous section would lead to optimization problems with tens of thousands of variables and constraints, which is incompatible with the intended use of the method in large-scale reduced-order modeling applications.

□ [JAHO: *No, the above excerpt is not accurate; I'd remove it.*] □

4.2 Standard relaxations and their limitations

A classical approach to address the intractability of zero-norm minimization problems consists in replacing the zero-norm by its **convex surrogate**, the ℓ_1 norm. This relaxation **promotes**

sparsity while leading to a convex optimization problem that can, in principle, be solved efficiently using modern optimization techniques.

This option was **explored in detail in the development of the SAW–ECM methodology**. While the ℓ_1 -based formulation does provide sparse solutions, it was found to be computationally expensive in practice due to the large dimensionality of the resulting optimization problems. In particular, the number of design variables and constraints required to enforce exactness over multiple subspaces made the approach unattractive for realistic problem sizes.

□ [JAHO: *I’d add that the number of points provided was not as low as the greedy approach eventually adopted; anyway, the above excerpt is not relevant, mainly because we wish to use a global methodology in the sequel (for smoothing purposes)*] □

In the present manifold-adaptive context, the situation is even more restrictive. The inclusion of **smoothness regularization** terms and the need to handle a large number of manifold sampling points would lead to convex optimization problems of **prohibitive size**, even when exploiting sparsity and structure. For this reason, ℓ_1 -based relaxations are not considered a viable alternative in the present work.

Another class of approaches relies on greedy algorithms, in which integration points are selected or discarded iteratively based on local criteria. Greedy strategies form the **computational backbone of the SAW–ECM methodology** and have proven to be effective in identifying sparse cubature rules shared across multiple subspaces. However, **greedy selection alone does not enforce any form of continuity** of the resulting weight vectors across different regimes.

In the manifold-adaptive setting, this limitation is critical. Since the ultimate objective of MAW–ECM is to construct a continuous mapping between latent coordinates and cubature weights through regression, discontinuous or highly irregular weight fields cannot be tolerated. As a result, greedy strategies in their original form are insufficient, as they do not provide any mechanism to control smoothness across the manifold.

4.3 Pruning as a feasible alternative

To overcome these limitations, we adopt a **pruning-based strategy** inspired by the **Continuous Empirical Cubature Method** previously introduced by the authors. While the objectives of the two approaches differ—the present problem seeks both the location of integration points and their manifold-dependent weights, whereas the Continuous Empirical Cubature Method focuses on continuous deformation of integration rules—the underlying philosophy is closely related.

□ [JAHO: *The above excerpt is not true ! Rephrase it*] □

The central idea is to **avoid attempting to solve the sparsification problem directly**. Instead, the process is **initialized from a feasible cubature rule**, obtained from the **standard ECM**, which satisfies **all exactness and positivity constraints across the manifold**. Starting from this reference configuration, **integration points are progressively removed in a controlled manner**.

At each **pruning step**, the integration point to be **removed** is selected based on a **global criterion**, such as the **magnitude of its associated weights averaged over all manifold sampling points**. Once a point is removed, the weights of the remaining integration points are recomputed so as to **restore feasibility of the exactness and volume constraints**. This redistribution step is formulated as an optimization problem that seeks to **minimally modify the existing weights** while preserving admissibility.

The **pruning process thus replaces the combinatorial sparsification problem by a sequence of tractable subproblems**, each of which operates within the feasible set defined by the constraints. By construction, **sparsity is increased monotonically as pruning proceeds**, while

feasibility is maintained at every stage. The introduction of smoothness-enhancing mechanisms, discussed in the previous section, ensures that the resulting weight fields remain continuous over the manifold, thereby enabling the subsequent construction of regression-based mappings between latent coordinates and cubature weights.

The detailed formulation of the pruning algorithm and the associated weight redistribution strategy is presented in the following section.

□ [JAHO: *This part should be rephrased to indicate that the sequence stops when, upon removing a given point, no feasible solution is found. Of course*] □

4.4 Exploiting the combinatorial structure through candidate sweeping

An additional aspect of the **sparsification problem** that can be exploited in practice is its **intrinsically combinatorial nature**. As emphasized in the development of the Continuous Empirical Cubature Method, the difficulty of the problem does not stem solely from the dimensionality of the design space, but also from the discrete choice of which integration points are retained or eliminated at each stage of the reduction process.

In the present manifold-adaptive setting, this observation leads to a **pragmatic and effective strategy when performing pruning**. At a given pruning step, an integration point is tentatively selected for elimination based on a global indicator of its relevance, such as the magnitude of its associated weights averaged over all manifold sampling points. Once this point is removed, a **weight redistribution problem** is solved in order to **restore feasibility**.

It may occur, however, that no feasible redistribution of the remaining weights exists for the selected candidate, for instance due to the activation of non-negativity constraints or the loss of sufficient degrees of freedom to satisfy the imposed conditions. In such cases, the **pruning attempt is declared unsuccessful for that particular integration point**.

Rather than terminating the reduction process or resorting to a global reformulation of the problem, the **combinatorial nature of the sparsification task can be further exploited** by considering **alternative candidates**. Specifically, the next **integration point in the ordered list**—typically the one with the next smallest average contribution to the constraints—is selected, and the redistribution problem is solved again. **This procedure is repeated until either a feasible solution is found or all remaining candidates have been tested**.

This candidate sweeping strategy, inspired by the Continuous Empirical Cubature Method, **transforms the sparsification problem into a sequence of feasibility checks over a finite and progressively shrinking set of discrete alternatives**. By combining global relevance indicators with local **feasibility restoration**, the method retains a high degree of robustness while avoiding the need for **exhaustive combinatorial searches**. Moreover, this approach naturally integrates with the pruning philosophy adopted in the present work, allowing sparsity to be increased incrementally while preserving feasibility and smoothness of the cubature weights across the manifold.

5 Optimization Problem for a Single Pruning Step

This section states the optimization problem solved at a single pruning step of the manifold-adaptive procedure. The purpose of this step is to eliminate one candidate integration point from the current cubature rule and to redistribute its contribution among the remaining points while preserving feasibility with respect to exactness, volume conservation, nonnegativity, and smoothness of the resulting weight fields over the manifold.

5.1 Setting and notation

Let $\mathcal{M} \subset \mathbb{R}^2$ denote the latent (master) coordinate domain, sampled at the set of manifold points

$$\{\boldsymbol{\xi}^{(j)}\}_{j=1}^{N_m}, \quad \boldsymbol{\xi}^{(j)} = (\xi_1^{(j)}, \xi_2^{(j)}),$$

which are assumed to form a rectangular grid in latent space. These samples are the nodes of a structured graph $\mathcal{G}$ used to encode smoothness of the weights over the manifold.

□ [JAHO: *This particularization is not necessary for explaining the method; I have introduce it just for clarity purposes.*] □

Let $\mathcal{I}_r \subseteq \mathcal{I}_0$ denote the index set of integration points currently retained at pruning iteration r , where $\mathcal{I}_0$ is the index set of the initial ECM rule and $|\mathcal{I}_r| = N_r$. At each manifold sample $\boldsymbol{\xi}^{(j)}$, the current cubature rule is represented by a nonnegative weight vector

$$\boldsymbol{\zeta}_r^{(j)} \in \mathbb{R}^{N_r}, \quad j = 1, \dots, N_m,$$

whose entries correspond to the integration points in $\mathcal{I}_r$.

For each manifold sample j , a set of n_c exactness constraints is prescribed. These are collected in a matrix $\mathbf{U}_r^{(j)} \in \mathbb{R}^{n_c \times N_r}$ and a right-hand side $\mathbf{b}^{(j)} \in \mathbb{R}^{n_c}$ such that

$$\mathbf{U}_r^{(j)T} \boldsymbol{\zeta}_r^{(j)} = \mathbf{b}^{(j)}.$$

In particular, one of these constraints enforces conservation of volume. Denoting by $\mathbf{1} \in \mathbb{R}^{N_r}$ the vector of ones, the volume constraint reads

$$\mathbf{1}^T \boldsymbol{\zeta}_r^{(j)} = V, \quad j = 1, \dots, N_m,$$

where V is the total integration volume.

5.2 Candidate elimination and feasible redistribution

At iteration r , a candidate integration point $p \in \mathcal{I}_r$ is selected for elimination. The elimination is enforced by requiring that the weights associated with p vanish at *all* manifold samples. Operationally, this is equivalent to constructing the reduced index set

$$\mathcal{I}_{r+1} = \mathcal{I}_r \setminus \{p\}, \quad |\mathcal{I}_{r+1}| = N_{r+1} = N_r - 1,$$

and seeking a new family of weight vectors

$$\boldsymbol{\zeta}_{r+1}^{(j)} \in \mathbb{R}^{N_{r+1}}, \quad j = 1, \dots, N_m,$$

defined only on the retained integration points $\mathcal{I}_{r+1}$.

The redistribution problem aims to determine $\{\boldsymbol{\zeta}_{r+1}^{(j)}\}_{j=1}^{N_m}$ such that the exactness constraints remain satisfied at every manifold sample, the weights remain nonnegative, and the new weights are close to the previous ones (restricted to $\mathcal{I}_{r+1}$).

□ [JAHO: *By exactness constraints, we mean the integration of the reduced, unassembled residual, and other conditions related to the evaluation of output of interests, such as reactions or homogenized stresses.*] □

5.3 Objectives: least-change and smoothness over the manifold

Two guiding principles govern the redistribution step.

Least-change with respect to the previous rule. Among all feasible redistributions after removing p , **preference is given to the new weights that differ as little as possible from the weights obtained at the previous iteration.** To state this precisely, let $\zeta_{r,\setminus p}^{(j)} \in \mathbb{R}^{N_{r+1}}$ denote the restriction of the previous weight vector $\zeta_r^{(j)}$ to the retained indices $\mathcal{I}_{r+1}$. The redistribution step seeks $\zeta_{r+1}^{(j)}$ close to $\zeta_{r,\setminus p}^{(j)}$ in a quadratic sense.

Smoothness of the weight fields over the manifold. Because MAW-ECM ultimately constructs a **regression map between latent coordinates and cubature weights**, the resulting weights must vary smoothly across the manifold sampling points. This requirement is enforced by **penalizing the lack of smoothness of each weight component over the manifold graph.**

Let $\mathbf{K} \in \mathbb{R}^{N_m \times N_m}$ be the graph Laplacian associated with the rectangular grid, interpreted as the conductance (stiffness) matrix of a diffusion operator discretized on the latent-domain mesh. For each retained integration point index $i \in \mathcal{I}_{r+1}$, define the vector collecting its weights over the manifold samples:

$$\zeta_{r+1,i} = \begin{bmatrix} \zeta_{r+1,i}^{(1)} & \zeta_{r+1,i}^{(2)} & \cdots & \zeta_{r+1,i}^{(N_m)} \end{bmatrix}^T \in \mathbb{R}^{N_m}.$$

Smoothness is promoted by penalizing the quadratic form $\zeta_{r+1,i}^T \mathbf{K} \zeta_{r+1,i}$ and summing this contribution over all retained integration points.

5.4 Single-step redistribution problem

The single pruning step is therefore posed as the following constrained optimization problem: find the manifold-dependent weight vectors $\{\zeta_{r+1}^{(j)}\}_{j=1}^{N_m}$ minimizing a combination of (i) a least-change term relative to the previous step and (ii) a manifold-smoothness penalty, subject to exactness, volume conservation, and nonnegativity:

$$\begin{aligned} \min_{\zeta_{r+1}^{(1)}, \dots, \zeta_{r+1}^{(N_m)}} & \frac{1}{2} \sum_{j=1}^{N_m} \left\| \zeta_{r+1}^{(j)} - \zeta_{r,\setminus p}^{(j)} \right\|_2^2 + \frac{\alpha}{2} \sum_{i \in \mathcal{I}_{r+1}} \zeta_{r+1,i}^T \mathbf{K} \zeta_{r+1,i} \\ \text{subject to} & \mathbf{U}_{r+1}^{(j)T} \zeta_{r+1}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \\ & \mathbf{1}^T \zeta_{r+1}^{(j)} = V, \quad j = 1, \dots, N_m, \\ & \zeta_{r+1}^{(j)} \geq \mathbf{0}, \quad j = 1, \dots, N_m. \end{aligned} \tag{8}$$

Here, $\alpha > 0$ controls the strength of the smoothness penalty. The matrices $\mathbf{U}_{r+1}^{(j)}$ denote the constraint operators restricted to the retained integration points $\mathcal{I}_{r+1}$.

Problem (8) constitutes the **formal statement of the redistribution step performed after tentatively eliminating a candidate integration point p** . In subsequent sections, we discuss practical strategies to compute feasible solutions efficiently, including the role of candidate sweeping when feasibility cannot be restored for a given elimination choice.

6 Global Formulation of the Single-Step Redistribution Problem

This section rewrites the single-step redistribution problem (8) in a global (monolithic) algebraic form. The purpose is to make explicit (i) the design variable (vector of unknowns), (ii) the quadratic objective in matrix form (including the “mass-like” and “stiffness-like” contributions), and (iii) the linear constraints assembled into a single global constraint matrix. This global view will later serve as a reference to motivate more practical solution strategies.

6.1 Design variables and size

At pruning step $r \rightarrow r+1$, the retained integration points define the index set

$$\mathcal{I}_{r+1} = \mathcal{I}_r \setminus \{p\}, \quad |\mathcal{I}_{r+1}| = N_{r+1}.$$

For each manifold sampling point $j = 1, \dots, N_m$, the unknown weight vector is

$$\boldsymbol{\zeta}_{r+1}^{(j)} \in \mathbb{R}^{N_{r+1}}.$$

We collect all unknowns into a single global design vector by stacking the weights sample-by-sample:

$$\mathbf{z} = \begin{bmatrix} \boldsymbol{\zeta}_{r+1}^{(1)} \\ \boldsymbol{\zeta}_{r+1}^{(2)} \\ \vdots \\ \boldsymbol{\zeta}_{r+1}^{(N_m)} \end{bmatrix} \in \mathbb{R}^{N_m N_{r+1}}. \quad (9)$$

Hence, the number of global unknowns is

$$N_{\text{dof}} = N_m N_{r+1}.$$

For reference, in the typical setting discussed in this work, $N_m \approx 700$ and $N_{r+1} \approx 80$ (early stages), yielding $N_{\text{dof}} \approx 5.6 \times 10^4$.

6.2 Quadratic objective: mass-like and smoothness terms

Least-change term. Let $\boldsymbol{\zeta}_{r,\setminus p}^{(j)} \in \mathbb{R}^{N_{r+1}}$ denote the restriction of the previous-step weights to the retained index set $\mathcal{I}_{r+1}$. Define the stacked reference vector

$$\mathbf{z}_{\text{ref}} = \begin{bmatrix} \boldsymbol{\zeta}_{r,\setminus p}^{(1)} \\ \boldsymbol{\zeta}_{r,\setminus p}^{(2)} \\ \vdots \\ \boldsymbol{\zeta}_{r,\setminus p}^{(N_m)} \end{bmatrix} \in \mathbb{R}^{N_m N_{r+1}}. \quad (10)$$

The least-change contribution in (8) can then be written compactly as

$$\frac{1}{2} \sum_{j=1}^{N_m} \left\| \boldsymbol{\zeta}_{r+1}^{(j)} - \boldsymbol{\zeta}_{r,\setminus p}^{(j)} \right\|_2^2 = \frac{1}{2} \|\mathbf{z} - \mathbf{z}_{\text{ref}}\|_2^2 = \frac{1}{2} (\mathbf{z} - \mathbf{z}_{\text{ref}})^T \mathbf{M} (\mathbf{z} - \mathbf{z}_{\text{ref}}), \quad (11)$$

where the matrix

$$\mathbf{M} = \mathbf{I}_{N_m N_{r+1}} \quad (12)$$

plays the role of a *mass-like* matrix. More general weighted least-change choices can be accommodated by letting $\mathbf{M}$ be any symmetric positive definite matrix (e.g. diagonal weights), but in this work the simplest form (12) is adopted.

Smoothness term. Let $\mathbf{K} \in \mathbb{R}^{N_m \times N_m}$ denote the graph Laplacian (conductance/stiffness matrix) associated with the rectangular grid in latent space. The smoothness penalty in (8) is defined componentwise across the manifold samples: for each retained integration point index $i \in \mathcal{I}_{r+1}$, the weights $\{\zeta_{r+1,i}^{(j)}\}_{j=1}^{N_m}$ form a vector in $\mathbb{R}^{N_m}$ whose roughness is measured by $\mathbf{K}$.

In global form, this term can be written using a Kronecker product. Specifically, the sum

$$\sum_{i \in \mathcal{I}_{r+1}} \zeta_{r+1,i}^T \mathbf{K} \zeta_{r+1,i}$$

is equivalent to

$$\mathbf{z}^T (\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}) \mathbf{z}, \quad (13)$$

where $\otimes$ denotes the Kronecker product. The matrix $\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}$ is sparse and symmetric, and acts as a *stiffness-like* coupling between neighboring manifold samples, applied identically to each of the N_{r+1} weight components.

Global quadratic objective. Combining (11) and (13), the objective in (8) can be written in standard quadratic form as

$$\min_{\mathbf{z}} \frac{1}{2} (\mathbf{z} - \mathbf{z}_{\text{ref}})^T \mathbf{M} (\mathbf{z} - \mathbf{z}_{\text{ref}}) + \frac{\alpha}{2} \mathbf{z}^T (\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}) \mathbf{z}, \quad (14)$$

with $\mathbf{M} = \mathbf{I}_{N_m N_{r+1}}$ unless stated otherwise.

6.3 Linear equality constraints: block structure

At each manifold sampling point j , the exactness constraints are written as

$$\mathbf{U}_{r+1}^{(j)T} \zeta_{r+1}^{(j)} = \mathbf{b}^{(j)}, \quad j = 1, \dots, N_m, \quad (15)$$

where $\mathbf{U}_{r+1}^{(j)} \in \mathbb{R}^{n_c \times N_{r+1}}$ and $\mathbf{b}^{(j)} \in \mathbb{R}^{n_c}$. The volume constraint is written as

$$\mathbf{1}^T \zeta_{r+1}^{(j)} = V, \quad j = 1, \dots, N_m. \quad (16)$$

Local orthogonality. In the present work, the constraint operators are assembled so that, at each manifold sample j , the rows of the combined local constraint matrix are orthogonal. To make this explicit, define the augmented local constraint matrix

$$\mathbf{A}^{(j)} = \begin{bmatrix} \mathbf{U}_{r+1}^{(j)T} \\ \mathbf{1}^T \end{bmatrix} \in \mathbb{R}^{(n_c+1) \times N_{r+1}}, \quad \mathbf{c}^{(j)} = \begin{bmatrix} \mathbf{b}^{(j)} \\ V \end{bmatrix} \in \mathbb{R}^{n_c+1}. \quad (17)$$

The orthogonality assumption means that $\mathbf{A}^{(j)} \mathbf{A}^{(j)T}$ is diagonal (or close to diagonal) for each j , which simplifies the numerical properties of local feasibility restorations discussed later.

Global equality constraints. Stacking (17) over all manifold samples yields the global linear system

$$\mathbf{A} \mathbf{z} = \mathbf{c}, \quad (18)$$

where

$$\mathbf{A} = \text{blkdiag}\left(\mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \dots, \mathbf{A}^{(N_m)}\right) \in \mathbb{R}^{N_m(n_c+1) \times N_m N_{r+1}}, \quad \mathbf{c} = \begin{bmatrix} \mathbf{c}^{(1)} \\ \mathbf{c}^{(2)} \\ \vdots \\ \mathbf{c}^{(N_m)} \end{bmatrix} \in \mathbb{R}^{N_m(n_c+1)}. \quad (19)$$

Therefore, as anticipated, the global equality constraint matrix $\mathbf{A}$ is *block diagonal* with respect to the manifold samples. In particular, the orthogonality property holds locally within each diagonal block, although it does not imply global orthogonality of $\mathbf{A}$ because the blocks correspond to distinct manifold points.

6.4 Positivity constraints in global form

Nonnegativity is enforced at every manifold sample:

$$\boldsymbol{\zeta}_{r+1}^{(j)} \geq \mathbf{0}, \quad j = 1, \dots, N_m.$$

In global notation, this becomes the componentwise inequality

$$\mathbf{z} \geq \mathbf{0}. \quad (20)$$

6.5 Compact global statement

Using (9)–(20), the single-step redistribution problem (8) can be stated compactly as

$$\begin{aligned} \min_{\mathbf{z} \in \mathbb{R}^{N_m N_{r+1}}} \quad & \frac{1}{2}(\mathbf{z} - \mathbf{z}_{\text{ref}})^T \mathbf{M}(\mathbf{z} - \mathbf{z}_{\text{ref}}) + \frac{\alpha}{2} \mathbf{z}^T (\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}) \mathbf{z} \\ \text{subject to} \quad & \mathbf{A}\mathbf{z} = \mathbf{c}, \\ & \mathbf{z} \geq \mathbf{0}, \end{aligned} \quad (21)$$

where $\mathbf{M} = \mathbf{I}_{N_m N_{r+1}}$ and $\mathbf{A}$ is block diagonal as in (19). The role of the smoothness term is to introduce coupling across manifold samples in the objective, while the constraint set remains separable across samples at the equality level, and separable componentwise at the inequality level.

6.6 Interpretation and Construction of the Smoothness Matrix $\mathbf{K}$

This subsection clarifies the role and structure of the matrix $\mathbf{K}$ introduced in the smoothness-augmented formulation, and provides a physical interpretation that guides its construction.

6.6.1 Physical interpretation

The physical intuition you describe is essentially correct.

The manifold sampling points $\{\boldsymbol{\xi}^{(j)}\}_{j=1}^{N_m}$ can be interpreted as the nodes of a finite element mesh defined over the latent-coordinate domain. In the present work, this domain is two-dimensional, and the sampling points form a rectangular grid, so the analogy with a structured finite element mesh is particularly natural.

Each node of this mesh does not carry a single scalar unknown, but rather a *collection* of scalar unknowns: one scalar weight per retained integration point. In other words, if N_{r+1} integration points are retained at a given pruning step, then at each manifold node there are N_{r+1} scalar fields

defined over the mesh. These scalar fields correspond to the spatial distribution, over the manifold, of the weights associated with each integration point.

From this viewpoint:

- the manifold nodes are the finite element nodes,
- the latent domain is the finite element domain,
- each cubature weight defines a scalar field over this domain,
- and smoothness of the weights corresponds to regularity of these scalar fields.

The matrix $\mathbf{K}$ is therefore interpreted as a *conductance* (or stiffness) matrix associated with a scalar diffusion problem posed on the manifold mesh. Importantly, this matrix acts only in the manifold direction; it does not mix different integration points. Each scalar weight field experiences the same diffusion operator.

6.6.2 Action of $\mathbf{K}$ on the global design vector

To make this explicit, recall that the global design vector is assembled as

$$\mathbf{z} = \begin{bmatrix} \zeta_{r+1}^{(1)} \\ \zeta_{r+1}^{(2)} \\ \vdots \\ \zeta_{r+1}^{(N_m)} \end{bmatrix} \in \mathbb{R}^{N_m N_{r+1}},$$

where each block $\zeta_{r+1}^{(j)} \in \mathbb{R}^{N_{r+1}}$ contains the weights at manifold node j .

The smoothness penalty is constructed so that, for each retained integration point index i , the vector

$$\zeta_{r+1,i} = \begin{bmatrix} \zeta_{r+1,i}^{(1)} & \zeta_{r+1,i}^{(2)} & \cdots & \zeta_{r+1,i}^{(N_m)} \end{bmatrix}^T$$

is treated as a scalar finite element field defined on the manifold mesh. The matrix $\mathbf{K}$ acts on this vector exactly as a standard stiffness or conductance matrix would act on nodal values of a scalar field.

There is therefore no coupling between different integration points at the level of $\mathbf{K}$. Coupling between integration points occurs only through the equality constraints (exactness and volume conservation), not through the smoothness operator.

This structure explains why the global smoothness contribution can be written as $\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}$: the same scalar diffusion operator is applied independently to each of the N_{r+1} scalar weight fields.

7 Handling Pruning without Reassembling the Global Operators

An important practical consideration in the global formulation presented above is the cost of repeatedly assembling the global matrices as pruning progresses. Since the pruning process may involve many elimination steps, rebuilding the mass-like matrix $\mathbf{M}$ and the smoothness matrix $\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}$ at every iteration would be unnecessarily expensive and conceptually redundant.

A key observation is that pruning an integration point does not change the geometry of the manifold, nor the graph structure used to define smoothness. What changes is only the *set of active degrees of freedom* associated with the cubature weights. This fact can be exploited to avoid reconstructing the global operators.

7.1 Physical analogy: elimination as constrained degrees of freedom

Recalling the physical interpretation introduced previously, the manifold sampling points are interpreted as nodes of a finite element mesh, and each retained integration point defines a scalar field over this mesh. Pruning an integration point corresponds to eliminating one of these scalar fields.

Equivalently, **from the point of view of the global design vector $\mathbf{z}$, pruning consists of enforcing that a subset of its components be identically zero.** In the physical analogy, this is analogous to **prescribing homogeneous Dirichlet conditions on certain degrees of freedom of a structure**, rather than removing the associated nodes or elements from the mesh.

Thus, eliminating an integration point does not require modifying the underlying conductance matrix $\mathbf{K}$ or the mass-like matrix $\mathbf{M}$. Instead, it amounts to **constraining the corresponding degrees of freedom to be zero.**

7.2 Encoding pruning through additional constraints

Let $\mathcal{I}_{r+1} \subset \mathcal{I}_0$ denote the set of integration points retained after pruning step $r + 1$, and let $\mathcal{E}_{r+1} = \mathcal{I}_0 \setminus \mathcal{I}_{r+1}$ denote the set of eliminated points. For any eliminated integration point index $i \in \mathcal{E}_{r+1}$, the pruning condition requires

$$\zeta_i^{(j)} = 0, \quad j = 1, \dots, N_m. \quad (22)$$

In global notation, these conditions correspond to linear equality constraints acting directly on the design vector $\mathbf{z}$. They can be appended to the global constraint system (18) as additional rows enforcing zero values for the corresponding components.

From this perspective, **pruning is handled by augmenting the constraint set, not by modifying the objective function or the global operators.** The matrices $\mathbf{M}$ and $\mathbf{K}$ remain fixed throughout the entire pruning process, while the **feasible set is progressively reduced by the addition of zero-value constraints.**

7.3 Advantages of a constraint-based pruning representation

Representing pruning through constraints has several advantages:

- The global operators $\mathbf{M}$ and $\mathbf{K}$ are assembled once and reused throughout all pruning steps.
- The geometric and smoothness properties encoded in $\mathbf{K}$ remain unchanged, ensuring consistent regularization across iterations.
- Pruning is treated in a unified manner together with other constraints (exactness, volume conservation, and nonnegativity), simplifying the conceptual structure of the problem.
- The progressive reduction of active degrees of freedom mirrors standard practices in constrained finite element analysis, where boundary conditions or contact constraints deactivate selected degrees of freedom without altering the underlying stiffness matrix.

This interpretation reinforces the view of pruning as a controlled restriction of the admissible solution space, rather than a modification of the underlying model. In subsequent sections, this perspective will be exploited to devise efficient solution strategies that leverage the fixed global structure while accommodating the evolving constraint set induced by pruning.

8 Solution Strategy Without Positivity Constraints: Elimination via Affine Boundary Conditions

This section discusses how the global redistribution problem (21) can be solved when the positivity constraints are temporarily ignored, i.e. when the optimization is subject only to linear equality constraints. The discussion adheres to the physical analogy introduced earlier and focuses on a solution strategy based on *elimination*, rather than on Lagrange multipliers.

8.1 Equality constraints as affine boundary conditions

In the physical analogy, the global design vector $\mathbf{z}$ plays the role of nodal degrees of freedom of a (multi-field) scalar diffusion-like problem posed on the manifold mesh. The smoothness term contributes a stiffness-like coupling, while the least-change term contributes a mass-like term.

The linear equality constraints

$$\mathbf{A}\mathbf{z} = \mathbf{c}$$

can be interpreted as *affine boundary conditions* (or, more generally, affine kinematic constraints). They define an affine subspace of admissible configurations: the sought minimizer is the equilibrium configuration (minimum energy state) *restricted* to this affine space.

In addition, pruning constraints of the type “these degrees of freedom must be zero” (arising from previously eliminated integration points, or from the candidate elimination enforced in the current step) are also linear equality constraints. In the same analogy, these correspond to homogeneous Dirichlet conditions on selected degrees of freedom. They can be treated on exactly the same footing as the exactness and volume constraints: all are linear equalities.

8.2 Why elimination is possible and natural here

You asked whether “we shall have a system of linear equations involving only the non-zero entries of our system.” Conceptually, you are right, with one important refinement:

- It is not the *entries of the solution* that are assumed non-zero in advance.
- Rather, it is the *degrees of freedom that remain free* after imposing equality constraints (including prescribed zeros) that define the reduced system.

In other words, elimination does not assume the solution is non-zero; it identifies which degrees of freedom are *prescribed* by constraints, and which remain *unknown*. The linear system is then written only for the unknown part.

This is exactly the classical finite element procedure for applying essential boundary conditions without Lagrange multipliers: one partitions the degrees of freedom into prescribed and free, substitutes the prescribed values, and solves for the free ones.

8.3 Quadratic objective in global matrix form

Ignoring positivity, the global problem (21) reduces to an equality-constrained convex quadratic minimization. Define the global Hessian (energy) matrix

$$\mathbf{H} = \mathbf{M} + \alpha (\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}), \quad (23)$$

and the reference vector $\mathbf{K} \leftarrow \mathbf{z}_{\text{ref}}$ as in (10).

□ [JAHO: *I'd simply write $(\mathbf{K} \otimes \mathbf{I}_{N_r+1})$*] □

Up to an irrelevant constant, the objective can be written as

$$\min_{\mathbf{z}} \frac{1}{2} \mathbf{z}^T \mathbf{H} \mathbf{z} - \mathbf{z}^T \mathbf{M} \mathbf{z}_{\text{ref}} \quad \text{subject to } \mathbf{A} \mathbf{z} = \mathbf{c}. \quad (24)$$

The matrix $\mathbf{H}$ is symmetric positive definite if $\mathbf{M}$ is positive definite (as in the present choice $\mathbf{M} = \mathbf{I}$), and $\mathbf{K}$ is positive semidefinite. This guarantees uniqueness of the minimizer over the feasible affine space.

8.4 Elimination through a feasible parametrization

The essential idea is to represent all vectors satisfying the equalities $\mathbf{A} \mathbf{z} = \mathbf{c}$ in a parameterized form that automatically enforces the constraints. This is the algebraic analogue of imposing affine boundary conditions by substitution.

A standard way to do this is as follows:

1. Compute (or characterize) one particular feasible vector $\mathbf{z}_p$ such that

$$\mathbf{A} \mathbf{z}_p = \mathbf{c}.$$

In physical terms, $\mathbf{z}_p$ is one admissible configuration satisfying the affine constraints.

2. Characterize the set of homogeneous admissible variations:

$$\mathcal{N}(\mathbf{A}) = \{\mathbf{v} : \mathbf{A} \mathbf{v} = \mathbf{0}\}.$$

□ [JAHO: *How to do that ? WHy not to use the approach described later, in Section 8.7 ?*] □

Any admissible solution can be written as

$$\mathbf{z} = \mathbf{z}_p + \mathbf{v}, \quad \mathbf{v} \in \mathcal{N}(\mathbf{A}).$$

3. Choose a basis $\mathbf{N}$ for $\mathcal{N}(\mathbf{A})$ so that any $\mathbf{v}$ can be written as $\mathbf{v} = \mathbf{N} \mathbf{y}$, where $\mathbf{y}$ is a vector of reduced unknowns. Then every feasible $\mathbf{z}$ is represented as

$$\mathbf{z} = \mathbf{z}_p + \mathbf{N} \mathbf{y}. \quad (25)$$

Substituting (25) into (24) transforms the constrained problem into an unconstrained quadratic minimization in $\mathbf{y}$ only. The optimality condition becomes a linear system involving the reduced matrix $\mathbf{N}^T \mathbf{H} \mathbf{N}$.

This strategy avoids Lagrange multipliers entirely and mirrors the elimination of constrained degrees of freedom in structural mechanics.

8.5 Incorporating prescribed zero entries

Constraints enforcing $\mathbf{z}_\ell = 0$ for certain indices ℓ (corresponding to eliminated integration points across all manifold samples) fit naturally into the same framework:

- If zeros are treated as explicit equality constraints, they are included as rows in $\mathbf{A} \mathbf{z} = \mathbf{c}$ with the corresponding entries in $\mathbf{c}$ equal to zero.

- Alternatively, and often more efficiently, these zero degrees of freedom can be eliminated *a priori* by removing them from the set of unknowns (standard Dirichlet elimination).

In the second option, one partitions the global vector into free and prescribed components,

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_f \\ \mathbf{z}_d \end{bmatrix}, \quad \mathbf{z}_d = \bar{\mathbf{z}}_d,$$

where $\mathbf{z}_d$ collects all degrees of freedom fixed by pruning (typically $\bar{\mathbf{z}}_d = \mathbf{0}$), and $\mathbf{z}_f$ collects the remaining free degrees of freedom. The quadratic form (24) and the constraint system $\mathbf{A}\mathbf{z} = \mathbf{c}$ can then be reduced by substitution, yielding a smaller problem in $\mathbf{z}_f$ only.

This is precisely the statement that, after enforcing prescribed zero entries, the final linear system is written only in terms of the remaining degrees of freedom. The important point is that “non-zero entries” should be understood as *free entries* (unknown degrees of freedom), not as entries that are guaranteed to be non-zero in the solution.

8.6 Resulting reduced linear system

After either (i) constructing a nullspace parametrization (25) or (ii) performing direct elimination of prescribed degrees of freedom and affine constraints, the solution of the equality-only problem reduces to solving a symmetric positive definite linear system in a reduced set of variables.

In mechanical terms:

- $\mathbf{H}$ plays the role of the global operator (mass + stiffness) governing the “energy” of the weight fields over the manifold.
- The equality constraints (including exactness, volume, and pruning zeros) act as affine kinematic constraints.
- Eliminating these constraints yields a reduced system on the unconstrained degrees of freedom, which can be solved efficiently by standard sparse linear algebra.

In the next section, we will exploit the specific block structure of the constraints in MAW-ECM (including local orthogonality of constraint rows and the block-diagonal structure of $\mathbf{A}$) to derive computationally efficient realizations of this elimination strategy.

8.7 Elimination via Linear Constraint Relations: Analogy with Periodic Boundary Conditions

The equality-only redistribution problem admits an alternative elimination strategy that closely mirrors the treatment of periodic (or multi-point) boundary conditions in finite element analysis. This viewpoint is particularly natural within the physical analogy adopted throughout this work and leads to an explicit reduced system of equations expressed directly in terms of the global Hessian.

8.7.1 Partition of degrees of freedom

Let the global design vector $\mathbf{z} \in \mathbb{R}^{N_m N_{r+1}}$ be partitioned into two subsets of degrees of freedom:

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_L \\ \mathbf{z}_R \end{bmatrix},$$

where:

- $\mathbf{z}_L$ collects the *independent* (or free) degrees of freedom,
- $\mathbf{z}_R$ collects the *dependent* (or constrained) degrees of freedom.

This partition is not necessarily related to spatial boundaries, but rather to the structure of the linear equality constraints arising from exactness conditions, volume conservation, and pruning-induced zero constraints. The choice of (L, R) is such that all equality constraints can be written as explicit linear relations expressing $\mathbf{z}_R$ as a function of $\mathbf{z}_L$.

□ [JAH0: *But how to carry out such a partition ?*] □

8.7.2 Affine constraint representation

Assume that the equality constraints can be rewritten in the affine form

$$\mathbf{z}_R = \mathbf{G} \mathbf{z}_L + \mathbf{u}, \quad (26)$$

where:

- $\mathbf{G}$ is a known matrix encoding the linear dependency between constrained and free degrees of freedom,
- $\mathbf{u}$ is a known vector accounting for non-homogeneous (affine) constraints.

This form is standard in the treatment of periodic boundary conditions, rigid links, and other multi-point constraints in finite element analysis. In the present context, $\mathbf{u}$ typically incorporates the right-hand sides of the exactness and volume constraints, while homogeneous constraints such as pruning-induced zeros correspond to rows of $\mathbf{G}$ with $\mathbf{u} = \mathbf{0}$.

8.7.3 Quadratic objective and Hessian partitioning

Ignoring positivity constraints, the redistribution problem is governed by the quadratic functional

$$\Pi(\mathbf{z}) = \frac{1}{2} \mathbf{z}^T \mathbf{H} \mathbf{z} - \mathbf{z}^T \mathbf{f},$$

where $\mathbf{H}$ is the global Hessian defined previously,

$$\mathbf{H} = \mathbf{M} + \alpha(\mathbf{K} \otimes \mathbf{I}_{N_{r+1}}),$$

and $\mathbf{f} = \mathbf{M} \mathbf{z}_{\text{ref}}$.

Partitioning $\mathbf{H}$ and $\mathbf{f}$ conformably with $\mathbf{z}$ gives

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{LL} & \mathbf{H}_{LR} \\ \mathbf{H}_{RL} & \mathbf{H}_{RR} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_L \\ \mathbf{f}_R \end{bmatrix}.$$

8.7.4 Reduced formulation by substitution

Substituting the affine constraint relation (26) into the functional $\Pi(\mathbf{z})$ yields an unconstrained quadratic functional in the free variables $\mathbf{z}_L$ alone. Differentiating with respect to $\mathbf{z}_L$ and enforcing stationarity leads to the reduced linear system

$$\mathbf{H}_{\text{red}} \mathbf{z}_L = \mathbf{f}_{\text{red}}, \quad (27)$$

where the reduced Hessian and right-hand side are given by

$$\mathbf{H}_{\text{red}} = \mathbf{H}_{LL} + \mathbf{H}_{LR}\mathbf{G} + \mathbf{G}^T\mathbf{H}_{RL} + \mathbf{G}^T\mathbf{H}_{RR}\mathbf{G}, \quad (28)$$

$$\mathbf{f}_{\text{red}} = \mathbf{f}_L + \mathbf{H}_{LR}\mathbf{u} + \mathbf{G}^T\mathbf{f}_R + \mathbf{G}^T\mathbf{H}_{RR}\mathbf{u}. \quad (29)$$

System (27) is symmetric and positive definite provided that $\mathbf{H}$ is positive definite on the admissible subspace, which is guaranteed in the present setting by the presence of the mass-like term.

Once $\mathbf{z}_L$ has been computed from (27), the constrained degrees of freedom are recovered by back-substitution using (26).

8.7.5 Interpretation

This formulation makes explicit the close analogy with periodic boundary conditions in finite element problems:

- $\mathbf{H}$ plays the role of the global stiffness (or stiffness-plus-mass) matrix,
- the equality constraints define linear relations between degrees of freedom,
- the matrix $\mathbf{G}$ encodes these relations,
- and the reduced system (27) governs the equilibrium of the unconstrained degrees of freedom.

In particular, no Lagrange multipliers are introduced, and all constraints are enforced exactly through elimination. This approach is especially attractive in the present context, as it allows the global operators $\mathbf{M}$ and $\mathbf{K}$ to remain unchanged throughout the pruning process, while only the constraint relations encoded in $\mathbf{G}$ and $\mathbf{u}$ evolve from one

8.7.6 Double-check: reduced system for $\mathbf{z}_R = \mathbf{G}\mathbf{z}_L + \mathbf{u}$

Consider the quadratic functional (positivity ignored)

$$\Pi(\mathbf{z}) = \frac{1}{2}\mathbf{z}^T\mathbf{H}\mathbf{z} - \mathbf{f}^T\mathbf{z},$$

with the partition

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_L \\ \mathbf{z}_R \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{LL} & \mathbf{H}_{LR} \\ \mathbf{H}_{RL} & \mathbf{H}_{RR} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_L \\ \mathbf{f}_R \end{bmatrix}.$$

Assume the affine constraint relation

$$\mathbf{z}_R = \mathbf{G}\mathbf{z}_L + \mathbf{u}.$$

Introduce the standard “constraint embedding” operator used in periodic/multi-point constraints:

$$\mathbf{T} = \begin{bmatrix} \mathbf{I} \\ \mathbf{G} \end{bmatrix}, \quad \bar{\mathbf{u}} = \begin{bmatrix} \mathbf{0} \\ \mathbf{u} \end{bmatrix}, \quad \text{so that } \mathbf{z} = \mathbf{T}\mathbf{z}_L + \bar{\mathbf{u}}.$$

Substituting into Π yields an unconstrained quadratic functional in $\mathbf{z}_L$, whose stationarity condition is

$$\mathbf{T}^T\mathbf{H}\mathbf{T}\mathbf{z}_L = \mathbf{T}^T\mathbf{f} - \mathbf{T}^T\mathbf{H}\bar{\mathbf{u}}.$$

Therefore the reduced linear system is

$$\mathbf{H}_{\text{red}}\mathbf{z}_L = \mathbf{f}_{\text{red}}, \quad (30)$$

with

$$\mathbf{H}_{\text{red}} = \mathbf{H}_{LL} + \mathbf{H}_{LR}\mathbf{G} + \mathbf{G}^T\mathbf{H}_{RL} + \mathbf{G}^T\mathbf{H}_{RR}\mathbf{G}, \quad (31)$$

$$\mathbf{f}_{\text{red}} = \mathbf{f}_L + \mathbf{G}^T\mathbf{f}_R - \mathbf{H}_{LR}\mathbf{u} - \mathbf{G}^T\mathbf{H}_{RR}\mathbf{u}. \quad (32)$$

Finally, the constrained degrees of freedom are recovered by back-substitution:

$$\mathbf{z}_R = \mathbf{G}\mathbf{z}_L + \mathbf{u}.$$

9 Handling Nonnegativity: Active-Set Enforcement and Practical Perspectives

The developments in the previous sections address the redistribution problem under linear equality constraints only. In practice, however, **pruning inevitably enters a regime in which the equality-only solution contains negative weights**. Since cubature weights must be nonnegative for physical and numerical reasons (e.g. stability, interpretability, and preservation of positive measures), nonnegativity must be enforced explicitly.

This section discusses practical ways to handle the appearance of negative entries, with emphasis on approaches that remain consistent with the elimination philosophy (affine boundary conditions) and avoid Lagrange multipliers whenever possible.

9.1 Why negativity appears and what it implies

When the positivity constraints are ignored, the redistribution problem is a **strictly convex quadratic minimization** over an **affine subspace defined by the equalities**. The unique minimizer in that subspace is therefore well-defined and easy to compute via elimination.

Negative entries arise because, after sufficient pruning, the remaining degrees of freedom are no longer rich enough to satisfy all equalities while keeping all weights nonnegative. In geometric terms, the affine subspace defined by the equalities intersects the nonnegative orthant in a set that may be:

- non-empty but increasingly “thin”,
- or empty for a given candidate elimination (infeasible pruning),
- or non-empty but requiring that some weights become exactly zero (active constraints).

Therefore, once negativity appears, the correct mathematical object is no longer an equality-constrained quadratic problem, but a **convex quadratic program with bound constraints**.

9.2 Active-set strategy: enforcing nonnegativity by iteratively fixing weights to zero

The central idea in active-set strategy is to treat nonnegativity constraints as *inequality boundary conditions*:

$$\mathbf{z} \geq \mathbf{0}.$$

When a component attempts to become negative, it is interpreted as an *active* constraint and the corresponding degree of freedom is clamped to zero, i.e. treated as a prescribed Dirichlet condition.

This leads naturally to an active-set iteration of the following form:

1. Solve the equality-only problem (using the elimination scheme described previously) to obtain a tentative $\mathbf{z}$.
2. Identify the set of indices where $\mathbf{z}$ is negative.
3. Add these indices to the active set and enforce $\mathbf{z}_\ell = 0$ for all active indices ℓ (i.e. impose additional homogeneous boundary conditions).
4. Resolve the equality-only problem with the enlarged set of prescribed zeros.
5. Iterate until no negative entries remain (or until infeasibility is detected).

This procedure reuses the **equality-only elimination machinery** almost verbatim: at each iteration, the only modification is that **additional degrees of freedom are transferred from the free set** to the constrained set, exactly as in contact problems where the contact set is updated.

Interpretation. In the physical analogy, the smoothness term penalizes gradients of the weight fields over the manifold, while the least-change term penalizes deviation from the previous state. The **equality constraints impose affine kinematic restrictions**, and the **active set introduces unilateral constraints analogous to a rigid obstacle at zero**. The resulting algorithm resembles **frictionless contact with a rigid foundation**, where the contact region (active set) is updated until consistency is reached.

9.3 Can the boundary-condition elimination scheme be reused?

Yes, with an important nuance.

The elimination scheme based on relations of the form $\mathbf{z}_R = \mathbf{G}\mathbf{z}_L + \mathbf{u}$ can still be used, but **it must be updated as the active set evolves**, because imposing $\mathbf{z}_\ell = 0$ moves degrees of freedom from the free set into the constrained set.

In practice, there are two equivalent ways to implement this reuse:

- **Direct Dirichlet elimination:** maintain an index set of fixed degrees of freedom (including pruned weights and active nonnegativity constraints) and eliminate them by substitution, solving at each iteration for the remaining free degrees of freedom.
- **Constraint relation update:** update the partition (L, R) and the relation $\mathbf{z}_R = \mathbf{G}\mathbf{z}_L + \mathbf{u}$ accordingly. In this view, each newly active nonnegativity constraint adds one more row to the “Dirichlet” part of the constraint relations.

Both perspectives avoid Lagrange multipliers. The first is typically simpler to implement because it does not require explicitly constructing $\mathbf{G}$; it only requires assembling the reduced system on the free indices.

9.4 Alternative perspectives: Lagrange multipliers and MATLAB toolbox solvers

Although the elimination-based active-set approach is consistent with your objective of avoiding Lagrange multipliers, it is useful to contrast it with other widely used strategies.

9.4.1 Lagrange multipliers (KKT systems)

If one enforces equality constraints using Lagrange multipliers, the equality-only problem leads to a saddle-point (KKT) linear system. When nonnegativity is included, one must additionally handle inequality constraints, typically via:

- active-set methods on top of the KKT system, or
- interior-point methods.

While mathematically elegant, this route tends to increase implementation complexity and can lead to indefinite linear systems. In large-scale settings, this may require specialized solvers and preconditioners. In your context, where the equality constraints are block diagonal and possess favorable local structure, elimination is often preferable.

9.4.2 Direct quadratic programming via MATLAB Optimization Toolbox

A different approach is to solve the full convex quadratic program directly using a generic solver (e.g. `quadprog`):

- *Pros*: robust off-the-shelf handling of bounds and equalities; active-set and interior-point options; less custom code.
- *Cons*: repeated solves at many pruning steps can be expensive; memory overhead; reduced ability to exploit the special structure (block-diagonal equalities and graph-coupled objective) unless carefully engineered.

For moderate problem sizes, `quadprog` may be entirely feasible. However, in a pruning workflow, one typically solves the redistribution problem many times, and therefore computational efficiency and reusability of factorizations become central. This is where custom elimination/active-set strategies can outperform generic solvers.

9.5 Recommended viewpoint for MAW–ECM

From the standpoint of MAW–ECM, the following perspective is often the most pragmatic:

- Treat the equality constraints (exactness and volume, plus pruned zeros) as affine boundary conditions and enforce them by elimination (as in periodic and multi-point constraints in FEM).
- Enforce nonnegativity through an active-set loop that progressively fixes negative weights to zero, thereby converting inequality constraints into additional homogeneous boundary conditions.
- Use candidate sweeping (discussed earlier) whenever a pruning attempt becomes infeasible due to excessive activation of constraints.

This approach is aligned with your current implementation philosophy: it preserves the physical analogy, avoids indefinite saddle-point systems, avoids reliance on monolithic global QP solves, and naturally accommodates the combinatorial nature of pruning.

In the following sections, we will specify concrete algorithmic realizations of this strategy, including stopping criteria, feasibility detection, and practical considerations for efficiently updating the set of constrained degrees of freedom across active-set iterations and pruning steps.

9.6 Convergence of the Active-Set Strategy under Volume Conservation

The convergence behavior of the active-set strategy in the present MAW-ECM setting must be interpreted in light of a key additional constraint: at every manifold sampling point, the sum of the cubature weights is required to be equal to the total integration volume. This constraint introduces a systematic bias in the redistribution process that strongly influences both the emergence of negative weights and the convergence of the active-set iteration.

9.6.1 Effect of the volume constraint on weight redistribution

The volume conservation constraint enforces that, for each manifold sample, the total mass represented by the cubature weights remains constant throughout the pruning process. As integration points are eliminated, the same total volume must be carried by a progressively smaller number of weights.

As a result, pruning induces a natural tendency for the remaining weights to increase on average. This effect is structural and independent of the particular optimization strategy adopted: it follows directly from the conservation of total volume. Consequently:

- negative weights do not arise because the algorithm “wants” to create negative mass,
- rather, they appear as a transient artifact of attempting to satisfy exactness constraints with insufficient degrees of freedom,
- while the volume constraint pushes the admissible solution toward higher positive values for the remaining weights.

This bias plays a stabilizing role in the active-set process, as explained next.

9.6.2 Why active-set convergence is favored in this context

The presence of the volume constraint significantly improves the convergence behavior of the active-set strategy compared to generic bound-constrained quadratic problems.

Monotone upward redistribution. When a negative weight is detected and clamped to zero, its contribution to the total volume must be absorbed by the remaining free weights. Since the volume constraint must still be satisfied exactly, this redistribution produces a net upward shift in the remaining weights. This makes it unlikely for already-positive weights to become negative in subsequent iterations.

Diminishing propensity for new activations. Because the remaining weights are collectively forced to grow, the set of negative entries tends to shrink rather than propagate. In practical terms, once a small set of weights is identified as incompatible with nonnegativity, fixing them to zero usually resolves the violations in a small number of iterations.

Compatibility with least-change regularization. The least-change term penalizes large deviations from the previous feasible weight field, which already satisfies nonnegativity and volume conservation. Together with the upward bias induced by the volume constraint, this term further discourages oscillatory behavior in which weights repeatedly cross zero.

9.6.3 Expected number of active-set iterations

Taking the volume constraint into account, one can refine the expectations stated earlier:

- In early and intermediate pruning stages, the equality-only solution typically violates nonnegativity in only a few components. The active-set strategy usually converges in a very small number of iterations (often one or two).
- As pruning becomes more aggressive, the number of required active constraints may increase, but the volume constraint continues to drive the remaining free weights upward, preventing widespread sign changes.
- Rapid growth of the active set is therefore a strong indicator of imminent infeasibility for the chosen candidate, rather than of slow convergence.

Thus, the iteration count is governed primarily by how many weights must be clamped to zero to restore feasibility, not by the total number of manifold samples or integration points.

9.6.4 Implications for predictability and failure detection

Because the sum of the weights is fixed at each manifold sample, the active-set process exhibits a form of built-in regularization that makes its behavior highly predictable in practice:

- convergence typically occurs after identifying a small, stable active set,
- failure manifests quickly, through rapid expansion of the active set or inability to satisfy the equality constraints once too many weights are fixed.

This observation reinforces the suitability of the active-set strategy for MAW–ECM: it exploits a structural property of the cubature problem (volume conservation) that naturally complements pruning and smoothness regularization. Combined with candidate sweeping, this leads to a robust and computationally efficient procedure for enforcing nonnegativity without resorting to Lagrange multipliers or generic quadratic programming solvers.

9.7 On the Maximum Achievable Reduction in the Number of Integration Points

It is instructive to discuss, at least at an idealized level, the maximum reduction in the number of integration points that can be achieved by the pruning strategy while maintaining feasibility of the constraints.

9.7.1 A fundamental lower bound imposed by local constraints

Consider a fixed pruning step in which, at each manifold sampling point, the weights must satisfy a set of linear equality constraints (exactness plus volume conservation). For simplicity of exposition, assume that the local constraint matrices are all of the same size, i.e. for every manifold sample j the augmented constraint system can be written as

$$\mathbf{A}^{(j)} \boldsymbol{\zeta}^{(j)} = \mathbf{c}^{(j)}, \quad \mathbf{A}^{(j)} \in \mathbb{R}^{m_{\text{loc}} \times N},$$

where m_{loc} is the number of local equality constraints, and N is the number of retained integration points at the current step.

A basic linear-algebra requirement for feasibility is that there exist a nonnegative vector $\zeta^{(j)} \geq \mathbf{0}$ satisfying the m_{loc} equalities. Regardless of the particular values in $\mathbf{A}^{(j)}$ and $\mathbf{c}^{(j)}$, one cannot in general expect to satisfy m_{loc} independent equalities with fewer than m_{loc} active unknowns. In other words, the smallest possible support size for a generic feasible solution is bounded below by m_{loc} .

From the perspective of the pruning process, this implies the following idealized bound:

If the local constraints at each manifold sampling point consist of m_{loc} independent equalities, then the number of retained integration points cannot, in general, be reduced below m_{loc} while preserving feasibility at every sampling point.

Therefore, starting from an initial ECM rule with N_0 integration points, the maximum achievable reduction is

$$N_0 - m_{\text{loc}},$$

and the minimum attainable number of integration points is m_{loc} .

If the number of constraints varies over the manifold, say $m_{\text{loc}}^{(j)}$ at sample j , then a conservative global lower bound is given by the worst-case scenario:

$$N_{\min} \geq \max_{j=1, \dots, N_m} m_{\text{loc}}^{(j)}.$$

This reflects the fact that the cubature rule must remain feasible simultaneously at all manifold samples.

9.7.2 Role of volume conservation and scaling of the remaining weights

The volume constraint strengthens the above argument by imposing a nontrivial normalization at each manifold point:

$$\sum_{i=1}^N \zeta_i^{(j)} = V.$$

If one approaches the limiting regime in which only m_{loc} integration points remain active, then the average magnitude of the surviving weights must necessarily scale as

$$\zeta_i^{(j)} \sim \mathcal{O}\left(\frac{V}{m_{\text{loc}}}\right),$$

simply because the total volume V must be distributed among only m_{loc} nonnegative entries.

This observation has two important implications for pruning:

- As the number of retained integration points approaches m_{loc} , the candidate weight chosen for elimination cannot be expected to be “small” anymore. The smallest removable weight (even on average) inevitably increases, because all remaining weights carry a significant fraction of the volume.
- Consequently, eliminating any further point requires a substantial redistribution of volume and exactness contributions among the surviving points, making feasibility restoration increasingly difficult.

9.7.3 Impact on active-set iterations and convergence difficulty

Near the limiting regime, the nonnegativity constraints become strongly binding. When a candidate point is removed, the required redistribution is large, and the equality-only solution is more likely to produce negative entries. The active-set procedure must then clamp multiple weights to zero and re-solve until nonnegativity is restored (or infeasibility is detected).

Therefore, one should expect that:

- the number of active-set updates per pruning step tends to increase as N approaches m_{loc} ,
- candidate sweeping is invoked more frequently, since some elimination choices will no longer admit any feasible redistribution,
- and the pruning process typically terminates naturally when no candidate can be removed without violating feasibility, which is expected to occur close to the bound $N \approx m_{\text{loc}}$ (or the worst-case $\max_j m_{\text{loc}}^{(j)}$ when non-uniform).

In summary, the number of local equality constraints provides a fundamental lower bound on the minimum attainable number of integration points, and the volume conservation constraint implies that, as this bound is approached, the magnitude of the remaining weights necessarily increases, making additional pruning progressively more challenging and convergence of the active-set strategy correspondingly more demanding.

9.8 Interpreting m_{loc} in Hyper-Reduced Manifold ROMs and Checking the Constraint Counts

In the present framework, the quantity m_{loc} denotes the number of *independent scalar equality constraints* that must be satisfied at *each* manifold sampling point. Equivalently, it is the number of scalar quantities that the reduced cubature rule must integrate exactly at that point (including the volume constraint and any additional macroscopic outputs that are enforced as constraints).

A useful way to reason about m_{loc} in hyper-reduced manifold ROMs is:

m_{loc} equals the dimension of the set of reduced equations you require to hold exactly at each manifold point, plus the number of additional exact macroscopic quantities you want to reproduce, plus the volume conservation constraint.

Residual constraints: tangent-space projection. In a manifold-based ROM, the reduced governing equations are often enforced by projecting the full residual onto the tangent space of the manifold. If the manifold is parameterized by d latent variables, the tangent space has dimension d . In that case, the number of residual-related scalar constraints is naturally

$$m_{\text{res}} = d.$$

This matches your statement: for $d = 2$, the projection of the residual onto the tangent space yields 2 scalar equations.

Volume constraint. Independently of the ROM enforcement, you impose at each sampling point the normalization

$$\sum_i \zeta_i^{(j)} = V,$$

which contributes exactly one scalar constraint:

$$m_{\text{vol}} = 1.$$

Macroscopic stress (or homogenized stress) constraints. If, in addition, you require the cubature rule to reproduce macroscopic stresses (or stress-like outputs) exactly at each manifold point, the number of scalar constraints equals the number of independent stress components of the chosen stress measure.

In homogenization of hyperelastic unit cells, it is common to consider the first Piola–Kirchhoff stress $\mathbf{P}$, which is generally non-symmetric. Therefore:

$$\dim(\mathbf{P}) = \begin{cases} 4, & \text{in 2D (a } 2 \times 2 \text{ tensor),} \\ 9, & \text{in 3D (a } 3 \times 3 \text{ tensor).} \end{cases}$$

If the constraints are formulated as exactness of the *averages* of these components, this contributes

$$m_{\text{stress}} = \begin{cases} 4, & \text{2D,} \\ 9, & \text{3D.} \end{cases}$$

(If instead one uses a symmetric stress measure in 2D or 3D, these counts would reduce to 3 and 6, respectively.)

9.8.1 Your 2D example: two latent variables (compression + shear)

Under your assumptions:

- Latent dimension: $d = 2 \Rightarrow m_{\text{res}} = 2$ (tangent-projected equilibrium).
- Volume constraint: $m_{\text{vol}} = 1$.
- Average stress constraints in 2D using $\mathbf{P} \in \mathbb{R}^{2 \times 2}$: $m_{\text{stress}} = 4$.

Hence,

$$m_{\text{loc}} = m_{\text{res}} + m_{\text{vol}} + m_{\text{stress}} = 2 + 1 + 4 = 7.$$

Your reckoning for the 2D case is therefore consistent, provided that the stress measure being constrained has four components (e.g. first Piola).

9.8.2 Your 3D example: hyperelastic homogenization with arbitrary loads

Your statement “6 latent variables” is consistent with parameterizing the macroscopic loading by a symmetric strain-like measure in 3D (six components). With that choice:

- Latent dimension: $d = 6 \Rightarrow m_{\text{res}} = 6$ (tangent-projected equilibrium).
- Volume constraint: $m_{\text{vol}} = 1$.
- Average stress constraints using $\mathbf{P} \in \mathbb{R}^{3 \times 3}$: $m_{\text{stress}} = 9$.

Hence,

$$m_{\text{loc}} = 6 + 1 + 9 = 16,$$

which matches your conclusion.

Remark on alternative conventions. If the latent variables were instead the full deformation gradient components ($d = 9$), and/or if one constrained a symmetric stress tensor (6 components), then m_{loc} would change accordingly. The counting rule remains the same; only the definition of what is enforced exactly changes.

9.8.3 Summary

In summary, your counts ($m_{\text{loc}} = 7$ in the 2D case and $m_{\text{loc}} = 16$ in the 3D case) are correct under the specific modelling choices you stated:

1. the reduced equilibrium constraints are the d tangent-projected residual equations,
2. one volume constraint is enforced at every manifold sample,
3. and the constrained macroscopic stress is taken as the full first Piola tensor (4 components in 2D, 9 in 3D).

10 Heuristics to Alleviate the Computational Burden

While the formulation developed in the previous sections provides a robust and systematic framework for manifold-adaptive pruning with smoothness and nonnegativity enforcement, its direct application at every pruning step may be computationally demanding. This section discusses a practical heuristic, supported by numerical experience, that significantly reduces the overall cost of the algorithm without compromising robustness.

10.1 A two-stage strategy: ignoring smoothness until it is needed

Numerical experiments indicate that a large fraction of the integration points can be eliminated without explicitly enforcing smoothness of the weights over the manifold. In particular, if the smoothness-promoting term involving the graph Laplacian is temporarily ignored, the redistribution problem simplifies considerably.

Specifically, if the objective is reduced to the least-change term alone, the single-step redistribution problem becomes:

- a strictly convex quadratic minimization with a diagonal mass-like matrix $\mathbf{M}$,
- subject only to the local equality constraints (exactness and volume),
- and temporarily ignoring both smoothness and nonnegativity.

Under these assumptions, the problem decouples across manifold sampling points. At each sampling point, one solves a small underdetermined linear system and selects, among the infinitely many feasible solutions, the one with minimum Euclidean norm relative to the previous weights. This is precisely the least-norm solution of an underdetermined system of equations.

10.2 Interpretation as local least-norm solves

Because $\mathbf{M}$ is diagonal (in fact, the identity in the simplest choice), the objective does not introduce any coupling across manifold samples. Consequently, the equality-only redistribution problem can be interpreted globally as a collection of independent local problems—one per manifold sampling point.

Each local problem consists of finding a weight vector of minimal norm that satisfies:

- the tangent-projected equilibrium constraints,
- the volume constraint,

- and the enforced zero entries associated with already pruned integration points.

This local structure allows the first stage of pruning to be implemented very efficiently, avoiding the construction and solution of large global systems. In particular, it eliminates the need to assemble or factorize matrices involving the graph Laplacian.

10.3 Why smoothness often emerges naturally

Provided that all weights produced by this least-norm solution remain nonnegative, numerical experience shows that the resulting weight fields are typically smooth across the manifold. This behavior can be understood intuitively:

- The least-change objective biases the solution toward the previous weight field, which is smooth by construction at earlier pruning steps.
- If the manifold on which the integrand lives is itself smooth, nearby sampling points tend to induce similar local constraint systems.
- As a result, the local least-norm solutions vary smoothly with the manifold coordinates, even in the absence of an explicit diffusion term.

In this regime, the smoothness-promoting term is effectively redundant and can be safely omitted, yielding substantial computational savings.

10.4 Triggering the transition to the full formulation

The limitation of the simplified approach becomes apparent when negative weights appear. In this situation:

- nonnegativity constraints become active,
- local least-norm solutions may develop discontinuities,
- and “speckled” or non-smooth weight distributions may emerge across the manifold.

This event serves as a natural trigger to transition from the simplified strategy to the full formulation developed in the previous sections. At that point, the smoothness term involving the graph Laplacian is reintroduced, and the global active-set strategy is applied to enforce nonnegativity and restore regularity of the weight fields.

10.5 Computational implications

This two-stage strategy has important computational consequences:

- Most pruning steps can be carried out using inexpensive local least-norm solves.
- The expensive global machinery (graph Laplacian, global active-set iterations) is activated only when strictly necessary.
- By the time the full formulation is required, the number of remaining integration points has already been drastically reduced.

As a result, the global systems solved in the later stages are significantly smaller than those that would arise if the full formulation were used from the outset. This dramatically reduces the total computational cost of the algorithm, while retaining the robustness and theoretical consistency of the manifold-adaptive pruning strategy.

In summary, this heuristic leverages the observation that smoothness is often an emergent property of least-norm redistribution on smooth manifolds, and that explicit regularization is only required near the feasibility boundary. It therefore provides an effective and principled means of accelerating the overall MAW–ECM procedure.

11 Contingency Strategies When the Global Redistribution Fails

Despite the robustness of the global redistribution strategy described above, there are situations in which the second-stage problem (including smoothness and nonnegativity) may fail to produce a feasible solution for a given candidate integration point. This section discusses contingency plans for such cases, focusing on localized strategies that reduce computational burden and increase robustness, as well as heuristics inspired by previous univariate implementations and related methods in other fields.

11.1 Failure modes of the global problem

Global failure typically manifests in one of the following ways:

- the active-set iteration grows rapidly, clamping a large number of weights to zero,
- the equality constraints become incompatible once too many degrees of freedom are fixed,
- or the resulting system becomes numerically ill-conditioned, indicating proximity to the feasibility boundary.

In all these cases, the failure is usually not global in nature, but rather localized in certain regions of the manifold where the constraints are hardest to satisfy after the candidate elimination.

11.2 Local redistribution on manifold neighborhoods

A natural contingency strategy is to replace the global redistribution problem by a set of *localized* problems defined on neighborhoods of the manifold graph.

Specifically, for a given manifold sampling point, one may consider a local patch consisting of that point and a small number of its neighbors (for example, the immediate neighbors in the rectangular grid, or a k -ring neighborhood). The redistribution problem is then solved only on this restricted subset of manifold points, enforcing:

- the local equality constraints on the patch,
- continuity or smoothness only within the patch,
- and compatibility with the previously computed weights outside the patch (treated as fixed boundary data).

This approach is analogous to domain decomposition or overlapping Schwarz methods in finite element analysis, where difficult global constraints are resolved through a sequence of local solves with information exchange at the interfaces.

The key advantages are:

- smaller problem sizes,
- improved numerical conditioning,
- and the ability to isolate problematic regions of the manifold.

11.3 Relation to your univariate heuristic

Your earlier univariate implementation follows a particularly simple version of this philosophy. By ignoring neighbors altogether and imposing a threshold on the number of negative weights allowed in a single redistribution step, you effectively restricted attention to purely local feasibility.

If the number of negative weights exceeded the threshold, the candidate was declared non-removable and the algorithm moved on to the next candidate. This strategy can be interpreted as an implicit locality criterion: only small, localized violations are tolerated and corrected.

This idea remains valid and meaningful in the multivariate manifold setting, where it can be generalized as follows:

- allow only a limited number of active nonnegativity constraints per manifold sample or per neighborhood,
- if this limit is exceeded, declare the current candidate infeasible without attempting full global redistribution.

11.4 Suggested hybrid heuristic: neighborhood-aware pruning

Based on experience with constrained mechanics, graph-regularized optimization, and multiscale methods, a promising hybrid heuristic can be articulated:

1. Attempt redistribution using the simplified first-stage approach (no smoothness).
2. If negative weights appear but are sparse, attempt the global smoothness-augmented redistribution.
3. If the global problem fails or becomes too expensive, identify clusters of manifold points where violations are concentrated.
4. Solve localized redistribution problems on neighborhoods around these clusters, keeping weights outside the neighborhoods fixed.
5. If local feasibility cannot be restored within a prescribed tolerance (e.g. maximum number of clamped weights or iterations), abandon the candidate and proceed to the next one.

This strategy combines the strengths of your original threshold-based heuristic with a more structured, graph-aware notion of locality.

11.5 Rationale and expected behavior

The rationale behind localized contingency strategies is that manifold-adaptive cubature rules typically fail first in regions where:

- the mapping between latent coordinates and responses is highly nonlinear,
- the effective dimensionality of the tangent space changes,

- or the exactness constraints become nearly linearly dependent.

By addressing feasibility locally, one avoids over-correcting regions of the manifold that are already well-behaved. Moreover, local failures are often strong indicators that the candidate integration point plays a critical role in satisfying constraints in that region, and therefore should not be removed.

11.6 Summary

In summary, when the global redistribution problem fails:

- local neighborhood-based solves provide a natural and physically motivated fallback,
- threshold-based heuristics on the number of negative weights remain effective and inexpensive,
- and hybrid strategies combining local and global reasoning offer a robust contingency plan.

These heuristics preserve the spirit of the MAW-ECM framework: pruning is driven not only by algebraic feasibility, but also by the geometric and local structure of the solution manifold.

12 Local Active-Set Strategies and Practical Simplifications

This section summarizes pragmatic strategies for enforcing nonnegativity during pruning while avoiding the solution of large global problems. The emphasis is deliberately placed on simplicity and robustness, as the present stage of the work is exploratory and aims primarily at validating the methodology. Computational optimality and fine-grained performance considerations are deferred to later revisions.

12.1 Avoiding global solves through locality

Although the redistribution problem admits a clean global formulation, experience shows that solving the full problem over all manifold sampling points is often unnecessary and, in difficult cases, counterproductive. Negative weights typically emerge in localized regions of the manifold, reflecting local incompatibilities between the reduced set of integration points and the imposed constraints.

This observation motivates a local strategy:

- Detect the manifold sampling points at which negative weights appear.
- Define a local patch around each such point, consisting of the point itself and a small neighborhood in the manifold graph (for instance, nearest neighbors in a rectangular grid).
- Solve a redistribution problem restricted to this patch, allowing weights to change only inside the patch while keeping weights outside fixed.

From the physical analogy viewpoint, this corresponds to solving a local equilibrium problem on a subdomain, with Dirichlet boundary conditions imposed by the surrounding manifold points. Smoothness is enforced only within the patch, which is sufficient to remove local irregularities without disturbing regions that are already well behaved.

If the local problem succeeds, the updated weights are reintegrated into the global weight field and pruning proceeds. If it fails, this is interpreted as evidence that the candidate integration point cannot be removed at the current stage.

12.2 Active-set enforcement of nonnegativity

Once negative weights appear, nonnegativity must be enforced explicitly. This is most naturally done using an active-set strategy, inspired by classical Nonnegative Least Squares methods.

The key principle is that clamping a weight to zero should not be viewed as a final correction, but as a modification of the set of admissible degrees of freedom. After clamping, the redistribution problem must be re-solved for all remaining free weights so that:

- the equality constraints remain satisfied,
- the least-change objective is respected,
- and the effect of the clamped weights is redistributed optimally.

In practice, this means that each time a set of negative weights is detected:

1. these weights are added to an active set and fixed to zero,
2. the redistribution problem is solved again for the remaining free weights,
3. the process is repeated until no negative weights remain or infeasibility is detected.

This strategy allows the remaining weights to adjust coherently, rather than freezing the solution after a blunt correction.

12.3 Clamping strategies: simplicity versus completeness

Several variants of active-set enforcement are possible, with different levels of sophistication.

Monotone clamping (simplest strategy). In this approach, once a weight is clamped to zero during a redistribution step, it remains clamped until that step is completed. No attempt is made to release previously clamped weights.

This strategy has several advantages:

- it is straightforward to implement,
- it avoids cycling of the active set,
- it is numerically stable in the presence of the volume constraint, which naturally pushes remaining weights upward.

Given the exploratory nature of the present work, monotone clamping is the preferred choice.

Reversible active sets (more complete strategy). A more elaborate alternative would allow weights to leave the active set if subsequent re-solves indicate that they should become positive again. This requires monitoring optimality conditions and updating the active set dynamically.

While potentially more accurate near degenerate configurations, this approach is significantly more complex and is not pursued here.

12.4 Rationale for the chosen level of complexity

The combination of:

- local redistribution problems,
- monotone active-set clamping,
- and candidate sweeping when infeasibility is encountered,

already provides a robust mechanism for handling nonnegativity without resorting to global quadratic programming solvers or Lagrange multipliers.

At this stage, the primary objective is to ensure that the algorithm behaves correctly, identifies removable integration points reliably, and produces smooth, nonnegative weight fields across the manifold. Once this behavior is firmly established, more refined strategies can be investigated to reduce computational cost or improve optimality.

In summary, the methodology intentionally favors simplicity and clarity over maximal generality, with the understanding that the proposed framework admits natural extensions as experience with the algorithm accumulates.

12.5 Local Nature of Nonnegativity Enforcement

It is important to clarify the scope at which nonnegativity constraints are enforced in the proposed MAW-ECM framework. In particular, a distinction must be made between *local* enforcement of nonnegativity over the manifold and *global* decisions related to pruning integration points.

A cubature weight in the present context is not a single scalar quantity. Instead, each integration point is associated with a *weight field* defined over the manifold of latent coordinates. Consequently, the appearance of a negative weight must always be interpreted with respect to a specific manifold sampling point.

When a negative value is detected, the correct interpretation is therefore the following:

A given integration point exhibits a negative weight at a specific manifold sampling point.

This does *not* imply that the same integration point must be clamped or removed at all other manifold samples.

Accordingly, nonnegativity is enforced *locally in manifold space*. When a negative entry is detected, the active-set mechanism clamps that particular weight to zero *only at the manifold sampling point where the violation occurs*. At other manifold points, the same integration point may remain free and strictly positive.

This local enforcement is consistent with the physical analogy adopted throughout the paper. The weight associated with a given integration point can be viewed as a scalar field defined over the manifold mesh. Imposing nonnegativity at a specific manifold node is analogous to prescribing a local Dirichlet condition at that node, rather than imposing a global restriction on the entire field.

It is also essential to distinguish local nonnegativity enforcement from the pruning decision itself. Pruning corresponds to a global operation in which an integration point is removed everywhere on the manifold, across all sampling points. This decision is based on aggregated measures of relevance (e.g. average weight magnitude or contribution to the constraints) and is independent of the active-set mechanism.

In contrast, active-set clamping operates at a finer level of granularity:

- it acts on individual entries of the weight field,
- it is local to specific manifold sampling points,
- and it serves only to restore feasibility of the redistribution problem.

This separation of roles is crucial for the robustness of the method. It allows integration points to remain active where they are needed, while enforcing nonnegativity only where it is violated. It also explains why local redistribution strategies and patch-based corrections are effective: negative weights typically appear in localized regions of the manifold and can be corrected without disturbing the remainder of the weight field.

In summary, the active-set strategy in MAW–ECM enforces nonnegativity locally in manifold space, while pruning decisions are taken globally. Confusing these two mechanisms would lead to overly aggressive elimination and loss of accuracy, whereas keeping them conceptually distinct enables flexible and stable reduction.

13 Candidate Selection and Smoothness Criteria in SAW–ECM Pruning

When extending empirical cubature pruning from a purely local formulation to a **manifold-adapted setting**, candidate selection can no longer be treated as a purely greedy process based solely on weight magnitude. **Different candidate integration points**, although seemingly equivalent from a local or averaged perspective, **may induce very different levels of smoothness** in the resulting manifold-indexed cubature rule. Since this smoothness is directly linked to the **regularity of the hyper-reduced operators** and, in particular, to the robustness of Jacobians in nonlinear problems, it must be explicitly accounted for during pruning.

□ [JAHO: *Robustness of Jacobians —I think it means robustness of the Newton-Raphson solution algorithm*] □

At each pruning iteration, the decision of which integration point to remove is therefore inherently **multi-objective**, balancing **feasibility**, **nonnegativity robustness**, **minimal redistribution of weights**, and smoothness across the manifold. In what follows, we formalize both global and purely local smoothness criteria and describe how they are combined into a practical candidate selection strategy.

13.1 Notation and setting

Let $\{\xi_j\}_{j=1}^{N_m}$ denote the sampled points on the solution manifold. At a given pruning iteration, let $\mathcal{I}$ be the set of active integration points, with $|\mathcal{I}| = N_r$. The manifold-indexed cubature weights are denoted by

$$\zeta = \left[\zeta_i^{(j)} \right] \in \mathbb{R}^{N_m \times N_r}, \quad j = 1, \dots, N_m, \quad i \in \mathcal{I}.$$

For a fixed integration point i , the vector

$$\zeta_i := \left(\zeta_i^{(1)}, \dots, \zeta_i^{(N_m)} \right)^\top$$

represents the variation of its weight over the manifold.

Let $\mathbf{L} \in \mathbb{R}^{N_m \times N_m}$ be the graph Laplacian associated with the manifold connectivity, constructed from the adjacency matrix of the manifold graph. This operator is the same as the one employed in the smoothness regularization of the SAW–ECM redistribution problem.

□ [JAHO: *L. I think this matrix was denoted by L in the foregoing...*] □

13.2 Smoothness-aware candidate evaluation

At each pruning step, candidate integration points are first preselected using a **classical importance indicator**, namely the average value of the weight across the manifold,

$$\mu(p) := \frac{1}{N_m} \sum_{j=1}^{N_m} \zeta_p^{(j)}, \quad p \in \mathcal{I}. \quad (33)$$

Rather than committing to the single smallest value of $\mu(p)$, a small subset of **top candidates** (e.g. the M smallest values) is retained for further testing.

For each candidate p in this **shortlist**, the pruning step is **tentatively applied and the constrained redistribution problem is solved**, yielding a **candidate-dependent weight field** $\zeta^{(p)}$.

Laplacian-based smoothness metric. A smoothness measure consistent with the SAW-ECM regularization is obtained from the quadratic form induced by the graph Laplacian. For each integration point $i \in \mathcal{I}$, define the column-wise smoothness energy

$$E_i := (\zeta_i)^\top \mathbf{L} \zeta_i, \quad (34)$$

and the total smoothness energy of the cubature rule as

$$E(\zeta) := \sum_{i \in \mathcal{I}} (\zeta_i)^\top \mathbf{L} \zeta_i = \text{trace}(\zeta^\top \mathbf{L} \zeta). \quad (35)$$

This quantity penalizes abrupt variations of the weights between neighboring manifold samples.

The impact of pruning candidate p on smoothness is quantified through the increment

$$\Delta E(p) := E(\zeta^{(p)}) - E(\zeta^{(\text{cur})}), \quad (36)$$

where $\zeta^{(\text{cur})}$ denotes the current weight field. Among feasible candidates, preference is given to those minimizing $\Delta E(p)$.

Worst-edge jump criterion. To explicitly detect localized discontinuities that may not be fully captured by $E(\zeta)$, a max-type edge-based metric is introduced. Let $\mathcal{E}$ denote the edge set of the manifold graph. Define

$$J_\infty(\zeta) := \max_{(a,b) \in \mathcal{E}} \|\zeta(a, :) - \zeta(b, :)\|_2, \quad (37)$$

$$J_\infty^{\max}(\zeta) := \max_{(a,b) \in \mathcal{E}} \max_{i \in \mathcal{I}} |\zeta_i^{(a)} - \zeta_i^{(b)}|. \quad (38)$$

Candidate comparison may then rely on the increment

$$\Delta J_\infty(p) := J_\infty(\zeta^{(p)}) - J_\infty(\zeta^{(\text{cur})}). \quad (39)$$

Consistency of the regularization parameter. All candidates tested at a given pruning iteration must be evaluated using the same value of the smoothing parameter α . Otherwise, smoothness metrics would not be comparable. Accordingly, α is held fixed during candidate testing and may only be adapted between pruning iterations.

13.3 Purely local smoothness and feasibility criteria

Prior to introducing manifold-level information, pruning decisions relied exclusively on local indicators. This logic is retained as a complementary feasibility criterion.

After removing a candidate integration point, a least-norm redistribution is first computed independently at each manifold sample under the equality constraints. Violations of nonnegativity are then corrected through an active-set mechanism. Let

$$\mathcal{A}^{(p)} \subset \{1, \dots, N_m\} \times \mathcal{I}$$

denote the set of clamped indices for candidate p . Define the clamping indicators

$$N_{\text{clamp}}(p) := \left| \mathcal{A}^{(p)} \right|, \quad (40)$$

$$N_{\text{clamp}}^{\max}(p) := \max_{j=1, \dots, N_m} \left| \left\{ i \in \mathcal{I} : (j, i) \in \mathcal{A}^{(p)} \right\} \right|. \quad (41)$$

Large values of these indicators signal that feasibility is achieved only through extensive active-set intervention and are therefore used to reject unsuitable candidates.

13.4 Combined selection strategy

Candidate selection proceeds hierarchically. First, candidates violating prescribed clamping thresholds are discarded. Among the remaining feasible candidates, global smoothness metrics are used to rank pruning options.

A convenient composite score is

$$S(p) := \mu(p) + \lambda_E \Delta E(p) + \lambda_J \Delta J_{\infty}(p) + \lambda_C N_{\text{clamp}}(p), \quad (42)$$

where λ_E , λ_J , and λ_C weight the relative importance of smoothness degradation, discontinuity detection, and nonnegativity pressure. This strategy preserves the interpretability and robustness of the original local pruning method while incorporating the manifold-aware smoothness required for stable SAW–ECM hyper-reduction.